Dice Duel

Rules of the Dice Duel

1. In each round, each player roles their die. Higher number wins a point.
   - If there is a tie, give each player the same amount. (Your choice: 0, 1/2, or 1 for the amount.)
   - *Why doesn't it matter what amount you award for a tie?*
2. Repeat for 36 rounds. Winner is the one with the most points.
3. In case of a tie, go into sudden death. First player to win a round wins the match.

Boring Dice Duel

4. If we use standard dice, the game is a little boring. Why?

Exciting Dice Duel

We can make dice duel more exciting by letting the players each create their own die, subject to these rules:

- Each side of the die must have a whole number from 0 to 6.
- The total of all the numbers must be 21.

Give some thought to what numbers you want to choose. The standard die has the numbers 123456.

4. Find a partner and play dice duel with your custom dice. Report the winner, the loser, and your dice.
5. We'll use this to start a kind of tournament. Each time you win, you move to the right on the score table. Once you have lost twice you are out.

Questions

6. Is there a best die? What would that even mean?
7. Is there a worst die? What would that mean?
8. How good is the standard die (123456)? Can you find a die that should beat it? Can you find a die that the standard die should beat? How can you tell?
9. Find a die that did well in our tournament. Can you find a die that should beat it? Can you find a die that should beat that die? If you keep going like this, will you find the best die?
10. By now, you should have realized that some dice have an advantage over some other dice. What is the largest advantage you can find between two dice?
11. If you could make your own die now, how would you make it? Does it matter if you know what die your opponent is using?
12. How many different dice are there?
13. What other questions could you ask about this game? Can you answer any of them?